

Chapter 13: The Equity Pillar in Practice — HEDIS Equity Measurement, HEROI, and the Science of Closing Disparities

The Analytical Toolkit for Health Equity Work: Stratified HEDIS Measurement, the HEROI Composite Score, Root Cause Analytics, Intervention Design by Disparity Type, and Vermont's Population Health Equity Architecture

From Identifying Disparities to Eliminating Them

Chapter 12 established the Vermont equity landscape — the geographic disparities that concentrate in the Northeast Kingdom, the racial and demographic disparities in coverage and access, the SDOH burden that compounds both, and the GLP-1 access crisis that represents the most consequential new equity issue in American healthcare. This chapter develops the analytical and operational toolkit for actually closing those gaps.

Health equity work that stops at measurement — producing disparity reports that sit on shelves while the disparities they document persist — has characterized most state-level equity efforts for the past two decades. The distinction between an equity report and an equity program is whether the measurement is connected to intervention, whether interventions are designed to address the specific root cause of each disparity, and whether there is an accountability structure that closes the loop between measurement, intervention, and outcome. Vermont's transformation mandate, with its explicit equity goals embedded in Acts 167 and 68 accountability frameworks, creates the institutional conditions for equity work that closes gaps rather than counting them.

Section 1: Stratified HEDIS Measurement — The Foundation of Equity Analytics

HEDIS — the Healthcare Effectiveness Data and Information Set — is the most widely used quality measurement system in American healthcare. But HEDIS in its standard form measures average performance across a plan or provider population. Average performance improvement can mask widening disparities when gains are concentrated in easier-to-reach, already-better-served populations. Equity requires HEDIS to be stratified — measured separately for subpopulations defined by race/ethnicity, income, geography, language, disability, and other equity-relevant dimensions.

The NCQA HEDIS Stratification Framework

The National Committee for Quality Assurance (NCQA) has developed HEDIS stratification standards that specify how plans and providers should measure and report quality performance by race, ethnicity, and other demographic dimensions. The HTR Health Equity Studio implements the NCQA stratification methodology against VHCURES and Blueprint data for Vermont providers. The following framework organizes the HEDIS measures most relevant to Vermont's equity priorities:

HEDIS domain	Key measures	Vermont equity stratification priority	Current Vermont data source
Access and preventive care	Adult BMI assessment; tobacco use screening and cessation; colorectal	Geographic (Northeast Kingdom vs. Burlington); income (Medicaid vs. commercial); race/ethnicity	VHCURES claims; Blueprint clinical registry for PCMH-attributed patients

	cancer screening; breast cancer screening; cervical cancer screening	(BIPOC vs. white Vermont adults)	
Chronic disease management	Hemoglobin A1c control for diabetics; blood pressure control for hypertensives; statin therapy for cardiovascular disease; medication adherence for chronic conditions	Geographic; income; BIPOC adults (Black and AIAN adults disproportionately affected by poorly controlled diabetes and hypertension in Vermont data)	Blueprint clinical registry (for PCMH-attributed patients); VHCURES pharmacy claims for medication adherence
Behavioral health	Follow-up after ED visit for mental illness (FUH); follow-up after ED visit for alcohol use (FUA); initiation and engagement of SUD treatment (IET); antidepressant medication management	Geographic (Rutland and Windham counties have statistically higher MH crisis rates); age (15-44); LGBTQ+ Vermonters (documented higher MH burden)	VHCURES; VDH ED surveillance data; Blueprint follow-up tracking
Maternal and reproductive health	Prenatal and postpartum care; well-child visits; adolescent well-care; prenatal depression screening	Income; race/ethnicity (BIPOC maternal outcomes nationally and in Vermont below white outcomes); geographic (rural access barriers to OB care)	VHCURES; Blueprint Women's Health Initiative data
Care coordination	30-day readmission rate; transitions of care; medication reconciliation post-discharge	Income (Medicaid patients have higher readmission rates nationally); geographic (distance from discharge follow-up resources)	VHCURES; GMCB hospital discharge data (VUHDDS)

Figure 13.1 — Vermont HEDIS equity stratification framework. Sources: NCQA HEDIS Technical Specifications; AHS Health Equity Data Report (January 2025); Blueprint for Health data.

The Data Collection Challenge — Getting Demographic Data Right

Stratified HEDIS measurement requires race/ethnicity and other demographic data at the individual patient level. This is where Vermont's equity measurement infrastructure has a specific gap: race/ethnicity data in VHCURES is incomplete. Claims data from commercial insurers typically lacks race/ethnicity fields; Medicare added race/ethnicity data collection more recently; Medicaid data has the most complete demographic information.

Vermont has two approaches to closing the race/ethnicity data gap: direct collection at the point of care (NCQA standards require plans and PCMH practices to collect and report patient demographic data) and indirect imputation using validated geocoding methods (the RAND-BISG method imputes race/ethnicity probability from last name and census block characteristics). Vermont's Blueprint for Health practices and FQHCs are the primary settings for direct demographic data collection; the coverage of this data for the broader Vermont population depends on VHCURES reporting requirements being strengthened.

Section 2: The HEROI Framework — A Composite Equity Score

The Health Equity Return on Investment (HEROI) framework is the HTR Health Equity Studio's proprietary approach to equity performance measurement for healthcare organizations operating in value-based care environments. HEROI addresses a specific problem in equity measurement: disparities exist across multiple dimensions simultaneously, and an organization that closes one disparity while widening another may show no improvement on any individual metric while its overall equity performance has not improved.

HEROI Components

The HEROI composite score is calculated across five equity dimensions, each weighted by its relative health impact and the strength of evidence for interventions that can address it:

- Access equity score (25% weight): Measures the gap in primary care, specialty care, and emergency care utilization between advantaged and disadvantaged subpopulations, stratified by race/ethnicity, income, and geography. Vermont focus: Northeast Kingdom vs. Burlington primary care access; BIPOC primary care provider relationship rates.
- Quality equity score (25% weight): Measures the gap in HEDIS quality measure performance between advantaged and disadvantaged subpopulations for the top 10 chronic disease management measures. Vermont focus: diabetes control, hypertension control, and preventive care rates stratified by income and race/ethnicity.
- Outcome equity score (25% weight): Measures the gap in utilization outcomes — avoidable hospitalizations, ED visits, 30-day readmissions — between subpopulations. Vermont focus: mental health and SUD crisis ED presentation rates by county; readmission rates by Medicaid vs. commercial payer status.
- SDOH burden score (15% weight): Measures the prevalence of health-related social needs in the organization's attributed population, stratified by demographics, and the completeness of SDOH screening and navigation. Vermont focus: HRSN screening completion rates by practice location; community resource referral completion rates.
- Trust and engagement score (10% weight): Measures patient experience across demographic subgroups using CAHPS survey data stratified by race/ethnicity and income. Vermont focus: CAHPS disparity scores for BIPOC Vermonters; patient-reported discrimination experiences in clinical settings.

A HEROI score of 80+ indicates an organization achieving equity across all five dimensions. A score of 60-80 indicates material disparities in one or more dimensions requiring targeted intervention. Below 60 indicates pervasive disparities that require comprehensive equity program redesign. Vermont's statewide HEROI score — calculated from VHCURES and Blueprint data — has not been formally computed; the analytics vendor procurement underway will enable this calculation.

Section 3: Root Cause Analytics — Matching Interventions to Causes

The most common failure in health equity intervention design is treating all disparities as if they have the same cause. A disparity in diabetes control rates between white and Black Vermont adults has a different root cause structure than a disparity in primary care access between Chittenden County and Essex County residents — and requires a different intervention. The five-category root cause taxonomy from Chapter 12 (geographic access, social needs, clinical practice variation, coverage barriers, trust and engagement) provides the structure for matching interventions to causes.

The Disparity Decomposition Method

The HTR disparity decomposition method breaks a measured disparity into its constituent causes by applying statistical decomposition techniques to population-level data. The method answers the question: of the observed gap in outcome X between group A and group B, how much is attributable to differences in coverage rates,

differences in primary care access, differences in clinical quality, and differences in SDOH burden?

For Vermont's diabetes control disparity — the 22% of Blueprint-attributed diabetic patients with HbA1c not under control, with statistically higher rates among Medicaid patients and rural patients — the decomposition analysis would estimate what share of the disparity is attributable to medication adherence failures (addressable through pharmacy access and MedReconciliation programs), care management gap (addressable through CoCM integration), SDOH burden (addressable through CHT navigation), and primary care access limitations (addressable through Blueprint expansion).

The practical value of decomposition is not statistical precision — it is intervention targeting. An equity program that invests in care management for a disparity driven primarily by medication access barriers will produce minimal results. An equity program that correctly identifies medication access as the primary driver and invests in pharmacy co-location, medication assistance programs, and automated refill reminders will close the gap efficiently.

The Five-Step Disparity Closure Process

The HTR Health Equity Studio implements the following five-step disparity closure process for client organizations:

Step	Action	Time investment	Vermont application
1. Baseline stratification	Calculate current HEROI score and identify the three largest measured disparities across all five equity dimensions	2-4 weeks with adequate data	VHCURES stratification analysis; Blueprint equity dashboard; VDH BRFSS county-level analysis for Northeast Kingdom vs. state average gaps
2. Root cause attribution	For each identified disparity, apply decomposition analysis to estimate the proportional contribution of each root cause category (access, social needs, clinical variation, coverage, trust)	4-6 weeks	Survey BIPOC patients and Northeast Kingdom residents to supplement claims-based decomposition; conduct focus groups with CHT staff about observed barriers
3. Intervention portfolio design	For each major root cause, select evidence-based interventions matched to root cause type and population. Match investment size to root cause contribution (don't fund transportation programs for a disparity primarily driven by clinical variation).	2-4 weeks	Vermont-specific: CCBHC for trust/access; CHW deployment for social needs; AI governance for clinical variation; BALANCE for coverage; HEART program for LGBTQ+ trust
4. Implementation with equity lens	Execute interventions with explicit equity targets: not 'increase primary care access' but 'increase Northeast Kingdom Medicaid patient primary care visit rate from 72% to 80% within 18 months'	Ongoing	Blueprint field staff set equity targets by HSA; AHS HSA Coordinators track progress; GMCB analytics vendor enables real-time monitoring
5. Accountability and iteration	Report disparity metrics alongside average metrics in all quality reporting. If a disparity is widening despite intervention,	Quarterly reporting cycle	AHS monthly transformation reports to include disparity-stratified metrics; HEROI score incorporated into AHEAD quality reporting for Vermont

	re-examine root cause attribution and intervention design.		
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Figure 13.2 — Five-step disparity closure process with Vermont applications. Source: HTR Health Equity Studio methodology.

Section 4: Equity in VBC Contracting — Protecting Against Unintended Consequences

Value-based care contracts have a well-documented potential to worsen health equity if not designed with explicit equity protections. The mechanisms are several: global budgets or capitation rates that do not adequately adjust for social risk may incentivize providers to avoid or under-serve high-SDOH populations; shared savings contracts that reward utilization reduction may reward avoiding care for complex patients; quality measure performance may be harder for organizations serving disadvantaged populations even when they are delivering excellent care relative to what their population can access.

Social Risk Adjustment — The Equity Design Requirement for VBC Contracts

Social risk adjustment — adjusting VBC benchmarks and performance targets to account for the social risk burden of an organization's attributed population — is the technical mechanism for protecting equity in payment design. An organization serving a population with high housing instability, food insecurity, and transportation barriers faces higher care management costs and lower quality measure performance potential than an organization serving a high-income, socially stable population. A shared savings contract that ignores this difference rewards the wrong organizations.

CMS has incorporated social risk adjustment into ACO REACH through Comprehensive and Professional Direct Contracting model designs that include social risk stratification. Vermont's AHEAD global budget design must incorporate social risk adjustment at the hospital level — ensuring that hospitals serving the Northeast Kingdom's high-SDOH populations are not penalized relative to Burlington-area hospitals with more favorable social risk profiles. This is an active design question in Vermont's AHEAD implementation that AHS and GMCB must resolve explicitly.

The VBC Equity Audit

Every VBC contract should undergo an equity audit before and after signing, examining whether the contract design creates incentives that could worsen disparities. The HTR VBC Equity Audit checklist covers eight categories:

- Attribution equity: Does the attribution methodology systematically exclude high-SDOH patients who lack a qualifying primary care visit?
- Social risk adjustment: Is the benchmark adjusted for the social risk profile of the attributed population?
- Quality measure equity: Are quality measures stratified by race/ethnicity and income, with performance reported and rewarded based on equity improvement, not just average improvement?
- SDOH investment incentives: Does the contract provide financial credit for investments in SDOH programs that reduce long-term utilization?
- Care management intensity: Are care management resources allocated proportionate to SDOH burden and disparity level, not just clinical risk?
- Beneficiary protections: Are there protections against provider avoidance of high-complexity, high-SDOH patients?
- Community benefit requirements: For global budgets, are there explicit requirements for community health investment, including SDOH programs?
- Equity performance reporting: Is equity performance reported publicly, with minimum standards below which the organization cannot be considered performing?

Section 5: Vermont's Population Health Equity Analytics Architecture

The analytical infrastructure for equity measurement in Vermont is being built through the convergence of three systems: VHCURES (claims-based population data), the Blueprint clinical registry (quality and demographic data for PCMH-attributed patients), and the VDH health equity data systems (BRFSS survey data, vital statistics, county-level environmental data). None of these systems alone is adequate for comprehensive equity measurement; their integration — the one health record per person mandate — is the target state.

In the interim, Vermont's equity analytics strategy must be pragmatic: use the best available data for each equity dimension, be explicit about gaps, and make decisions based on the evidence available while investing in improving the evidence base. The following architecture organizes Vermont's current equity analytics capabilities and the priority investments needed to close gaps:

Equity dimension	Best current data source in Vermont
Geographic access disparities (HSA-level)	VHCURES county-level utilization; GMCB hospital market share data; Blueprint practice location mapping
Insurance coverage disparities	KFF/ACS coverage estimates; BRFSS insurance status by demographic group; DVHA Medicaid enrollment data
Primary care access disparities (race/ethnicity)	VDH BRFSS survey (limited by small BIPOC sample size); Blueprint provider relationship data for attributed population
Chronic disease control disparities	Blueprint PCMH clinical registry (most complete, but only covers PCMH-attributed patients); BRFSS self-reported chronic disease by demographic group
Behavioral health disparities (geographic)	AHS ED surveillance data (suicide, SH, OD rates by county — available and stratified)
SDOH burden distribution	Blueprint HRSN screening data (incomplete — not all practices at full adoption); VDH community vulnerability index; USDA food access data by census tract
Clinical practice variation by demographic	VHCURES claims (limited demographic data); VDH health equity data — requires improved race/ethnicity data collection in clinical settings
Trust and patient experience	CAHPS survey data from Blueprint PCMH practices (available but limited demographic stratification)

Figure 13.3 — Vermont equity analytics: current data sources by equity dimension. Sources: GMCB; VDH; Blueprint for Health; DVHA; AHS.

Key Concepts

Term	Definition
HEDIS stratification	The measurement of HEDIS quality indicators separately for subpopulations defined by race/ethnicity, income, geography, and other equity-relevant dimensions. Stratification reveals disparities hidden in aggregate average performance data.
HEROI (Health Equity Return on Investment)	The HTR Health Equity Studio's composite equity performance score, calculated across five dimensions:

	access equity, quality equity, outcome equity, SDOH burden, and trust and engagement. Enables organizations to track overall equity performance across dimensions simultaneously.
Disparity decomposition	A statistical technique that separates an observed health disparity into its constituent root causes — coverage differences, access differences, clinical quality differences, SDOH burden differences — enabling targeted intervention design.
Social risk adjustment	Modification of VBC benchmarks and performance targets to account for the social risk burden of an organization's attributed population, preventing the penalization of providers serving high-SDOH populations.
CAHPS (Consumer Assessment of Healthcare Providers and Systems)	The standard patient experience survey used across Medicare, Medicaid, and commercial healthcare programs. When stratified by race/ethnicity and income, CAHPS data reveals differential patient experience quality across demographic groups.
VBC equity audit	A structured review of a VBC contract design for provisions that could worsen health disparities — inadequate social risk adjustment, attribution methodology that excludes high-SDOH patients, quality measures not stratified by equity dimensions.
BISG imputation	Bayesian Improved Surname Geocoding — a validated method for imputing race/ethnicity probability from patient last names and residential census block characteristics, used to supplement incomplete race/ethnicity data in claims databases.

Sources: NCQA HEDIS Equity Technical Specifications; HTR Health Equity Studio methodology; Vermont Department of Health Health Equity Data Report (January 2025); Act 167 of 2022; Act 68 of 2025; Blueprint for Health Annual Report (2024); KFF; VHCURES documentation; AHS Transformation Reports (2025).